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Task 1. Multimedia Network for Audio video streaming

Core concept:

Implement IP Multicast.

Aim:

To design and configure a multimedia network capable of streaming audio and video to multiple end devices.

Software Required:

- Cisco packet tracer

Network topology components

1 Server: configured as the multimedia / streaming server.

1 Switch: Cisco catalyst 2960 to connect the devices

3 End Devices: PCs(or) Laptops to act as stream clients

Straight through cables: to interconnect all devices

Procedure

Step 1: Build the physical topology

1. Drag and drop one Server into the workspace

2. Drag and drop one switch (2960-24TT).

3. Drag and drop three PCs (PC0, PC1, PC2).

4. Connect each PC and the server to the switch using copper straight through cables.

Step 2. configure IP Addresses

- Server IP 192.168.1.10, subnet mask 255.255.255.0

- PC0: IP 192.168.1.1, Subnet mask 255.255.255.0

- PC2 IP 192.168.1.3, subnet mask 255.255.255.0.

Step 3. enable and configure the multimedia Service.

1. click on the server, go to the services tab

2. Select HTTP/HTTPS

3. Ensure HTTP is on

4. Edit the index.html file to include simulated links for audio and video streams

Step 4:

Verify Multicast / simultaneous traffic

1. Switch from Real time mode to simulation mode in the bottom-right corner.

2. Edit filters to show only HTTP and TCP traffic:

3. Open the web browser desktop application on PC0, PC1 and PC2.

4. Enter URL http://192.168.1.10 on all three PCs and hit Go.

Sample output & verification

- Ping test: successful ICMP pings from any PC to the server.

- Web browser verification: each PC successfully loads the server's webpage displaying ~~on~~ the multimedia links.

- Traffic flow:
In simulation mode, the switch concurrently routes web traffic requests from all three client PCs to the streaming server and returns packets simultaneously

Result:
The multimedia network was successfully designed and configured in Cisco packet tracer. Simultaneous audio/video content delivery was verified using concurrent browser request from multiple client devices

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2. AAA security in an IOT environment

Aim:-
to set up local and server based AAA security methods for IOT devices using cisco packet tracer.

Procedure:-

1. Add an IOT home gateway, an AAA / RADIUS server, an Admin PC, and smart devices.

2. For local AAA : configure login credential directly inside the IOT gateway settings.

3. for service AAA : Activate the AAA service on the server, select radius, create user profiles and link gateway.

4. connect the Admin PC to the gateway (or) server to test credential enforcement when accessing IOT controls.

Result:-
Local and server based AAA security methods were successfully established to secure access to the IOT environment

3. Wireshark packet Analysis

Aim:

To capture and analyse IP packet header fields and payloads using wireshark.

Procedure

* Open wireshark software
* Select active network interface
* Start packet capture live.
* Generate HTTP (or) ping traffic
* Stop packet capture
* Type ip in filter bar.
* Select one IP packet
* Expand internet protocol version 4 layer
* Analyze Version, TTL and addresses
* Inspect data payload segment below.

Result:-

* IP packets captured successfully
* Header fields and payloads analysed.

4. Basic Network setup

Aim: To configure a functional local area network consisting of three PCs and two switches to ensure seamless communication across all devices.

Procedure.

The setup begin by placing three PCs and two network switches into the workspace within cisco packet tracer. Connect PC1 and PC2 directly to ports on Switch 1 and connect PC3 to a port on Switch 2 using standard ethernet straight through cables. Next, establish a connection between switch 1 and swith 2 using an ethernet crossover cable to link the two segments together. Once the physical topology is connected manually assign unique static IP addresses within the same subnet. finally, verify the end to end network connectivity by opening the command prompt on PC 1 and successfully executing a ping command to the IP addresses of PC 2 and PC 3.

Result: the network topology is successful established and verified, demonstrating active link lights and error-free ping replies which confirms that all three PCs can reliably communicat with each other across both switches